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Demographic predictors of public speaking anxiety among university students

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Abstract
Public speaking is a crucial component of many higher education courses and is essential for students' academic performance and future career success. Despite its importance, public speaking anxiety is a common issue among higher education students, adversely impacting their learning. Addressing this anxiety through targeted interventions, especially for the most at-risk students, is vital. This study provides insights into the demographic predictors of public speaking anxiety, using a sample of 1745 students from a large public university in the Czech Republic. We employed the Personal Report of Confidence as a Speaker in a short form to assess public speaking anxiety levels. Our multivariate regression analysis identified gender, type of high school, and study level as significant predictors of public speaking anxiety, whereas age, nationality, and field of study were not. The study found that women, non-binary individuals, graduates of academic high schools, and bachelor's students are more prone to public speaking anxiety. These findings highlight the need for targeted intervention and support strategies for students with higher levels of public speaking anxiety.

Keywords Public speaking anxiety · Communication apprehension · Speech anxiety · Speaking fear · Social anxiety · Mental health

Introduction

Public Speaking Anxiety (PSA) is among the most widely faced anxieties among higher education students. Many higher education courses require students to give a speech or a presentation in front of their class as part of the assessment (Farris et al., 2013; Grieve et al., 2021). However, talking in front of many people is often a major cause of anxiety for students. In a nationally representative sample of Finnish university students, public speaking and presentation situations were reported to lead to the highest levels of anxiety among all learning situations among students (Pörhölä et al., 2019). In the US, 61% of college students were found to be fearful of public speaking, ranking second behind the fear of death (Dwyer & Davidson, 2012). A survey of students from two UK universities provided similar results, with 80% of students reporting anxiety from oral presentations (Russell & Topham, 2012). In a pilot study at Murdoch University comprising 16 students, half of them scored in the range of clinically significant anxiety (Martin-Lynch et al., 2016).

PSA refers to a pronounced fear or nervousness about speaking in front of an audience. It typically involves a concern about being judged or negatively evaluated by others, leading to feelings of discomfort, apprehension, or avoidance behaviors when faced with public speaking situations. It is a subtype of social anxiety, manifested as physiological, cognitive, and behavioral responding, including shaking, perspiration, increased heart rate, depressive thoughts, panic symptoms, and avoidance of stressor situations (Bodie, 2010).

PSA has negative consequences for both learning and later job success of higher education students. PSA impairs memory (Sawyer & Behnke, 1997) and depletes positive academic emotions, including motivation and engagement, making learning less effective (Pekrun et al., 2002). Public speaking is perceived by both students and alumni as one of the most important skills they should acquire (Farris et al., 2013) and a skill crucial for their job success (Johnson &

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Szczupakiewicz, 1987; Smith & Sodano, 2011; van Ginkel et al., 2019). Indeed, public speaking is an important skill for many job positions, and PSA was found to negatively influence career expectations of higher education students and to be related to students preferring back-office careers (Buser & Yuan, 2023).

Considering the widespread presence of PSA among higher education students and the importance of public speaking on students' learning and careers, it is important to minimize the impact of PSA on student outcomes. An older meta-analysis conducted by Allen et al. (1989) revealed the effectiveness of various treatments for PSA, including cognitive modification, systematic desensitization, and skills training. Notably, the combination of all three treatments yielded the most substantial positive effects. In recent years, new approaches have emerged, offering cost-effective interventions through Internet-based self-help programs (Tillfors et al., 2008) and virtual reality (Lindner et al., 2019; Takac et al., 2019). The advent of these interventions has significantly reduced the overall cost of addressing PSA. Lim et al.'s (2023) meta-analysis underscores the efficacy of virtual reality in mitigating PSA. However, it also emphasizes the importance of integrating these novel approaches with traditional treatments rather than supplanting them entirely. Thus, despite the availability of affordable intervention programs, the vital component lies in complementing them with established treatments. It is therefore important to provide targeted and delivered intervention to the most at-risk student populations, underscoring the importance of examining the relationship between demographic characteristics of higher education students and their PSA.

Demographic predictors of PSA

Age

As people age, their emotional regulation abilities tend to improve (Brummer et al., 2014; Santos et al., 2021). Therefore, it can be expected that older students would exhibit lower PSA due to their enhanced capacity to utilize effective coping mechanisms and maintain emotional stability in stressful situations. This thesis was supported by a study on general population, which found that after 18th year, there was a sharp decline in PSA, with a slight increase later in life (Caballo et al., 2008). On the other hand, several studies on student populations did not find age to have any effect on students' PSA levels (Marinho et al., 2017, 2019; Phillips et al., 1997).

Gender

The stereotype threat theory posits that individuals from groups targeted by negative stereotypes may be more inclined to embody those stereotypes. This theory is particularly pertinent to women, who frequently face doubts about their competence in various domains, including assertive, public roles (Laguía et al., 2022; Neal-Jackson, 2020; Seo & Lee, 2021). Awareness of these stereotypes, especially when combined with the heightened visibility of public speaking, can undermine women's performance and exacerbate their PSA. Empirical evidence supports this notion, showing that women tend to report higher levels of PSA (Caballo et al., 2008; Hunter et al., 2014; Marinho et al., 2017; Perveen et al., 2018), demonstrate a reluctance to engage in public presentations (De Paola et al., 2021), and exhibit stronger physiological responses to public speaking (Carrillo et al., 2001) compared to men. Conversely, some studies have found no significant gender differences in self-reported PSA levels, in other studies, women and men were found to have the same self-reported PSA levels (Marinho et al., 2019; Phillips et al., 1997), and in one study, men were found to have higher PSA levels than women (Rodero & Larrea, 2022). None of the studies looked at PSA among non-binary students. However, research on general anxiety levels among non-binary people suggests that they experience greater anxiety levels compared to binary people (Butler et al., 2019; Thorne et al., 2020).

Nationality

To our knowledge, no study has looked at the influence of students' status as foreign students on their level of PSA. Yet, it may have a threefold influence on PSA. First, there is a probability that for foreign students the language of instruction is not their first language, with students' level of language ability predicting their PSA (Kelsen, 2019). Second, foreign students generally have higher anxiety levels stemming from their relocation and adjustment (Khoshlessan & Das, 2019; Wang et al., 2023). Third, similarly to gender, foreign students often encounter negative stereotypes regarding their abilities (Quinton, 2019; Spencer-Rodgers, 2001), which could in turn exacerbate their PSA.

High school type

Based on Rachman's (1980) theory of emotional processing, it can be expected that repeated exposure to public speaking would facilitate reduction of PSA. The type of high school may predict students' PSA levels due to different exposure levels of public speaking in curricula of vocational and academic high schools. Academic high schools

usually put higher emphasis on developing communication and public speaking skills than vocational high schools. In the Czech Republic, academic high schools provide more lessons focused on developing communication skills than vocational high schools (Ministry of Education, Youth and Sports of the Czech Republic, 2024). Hence, academic high school graduates – after having more exposure to public speaking – may experience lower levels of PSA in higher education.

Study level

Similarly to the high school type, the level of study may predict students' PSA levels due to different levels of exposure to public speaking, with students at higher levels having been exposed to public speaking more during their past academic experience. To this date, no study has looked at the influence of study level on students' level of PSA.

Study field

Field of study can be a predictor of students' PSA if certain fields offer more exposure to public speaking than others. Humanities and social sciences often emphasize argumentation and presentation as integral components of the curriculum, potentially providing students with more opportunities to develop comfort and proficiency in public speaking. This contrasts with the natural sciences, where the curriculum may prioritize written communication and technical proficiency, possibly resulting in less frequent engagement with public speaking tasks. So far, a single study looked at the effect of the field of study on students' PSA levels – finding no effect (Marinho et al., 2017).

Present study

While PSA among higher education students is widespread, research on demographic predictors of PSA is limited and the findings are mixed. Previous research on demographic predictors of PSA among higher education students does not provide a coherent picture of which groups of students are most at-risk of having PSA. Findings on gender are mixed, research on non-binary students is nonexistent, and several theoretically important demographic characteristics have not been examined.

In the present study, we aim to fill the research gap by investigating six demographic predictors of PSA among higher education students – age, gender, nationality, high school type, study level, and field of study – aiming to provide a clearer picture of who the most at-risk of PSA students are. We work with a rich sample of students from a single university in the Czech Republic. Our study provides a significant contribution to the existing body of knowledge on PSA, offering practical implications for educators, administrators, and mental health professionals working within academic settings. We address the question "How do demographic factors influence PSA among higher education students?" Based on the outlined theories and previous research, we test the following hypotheses:

H1: PSA decreases with age.
H2: PSA is higher among women and non-binary students.
H3: PSA is higher among non-Czech students.
H4: PSA is higher among vocational high-school graduates.
H5: PSA decreases with high study levels.
H6: PSA is higher among natural sciences students.

Methods

Instrument

We developed an online questionnaire consisting of several demographic items and items to assess students' level of PSA. The demographic items consisted of questions on age (continuous scale), gender (categorical: man, woman, non-binary), nationality (open-ended), type of high school from which they graduated (categorical: vocational/academic), study level (categorical: bachelor's, master's, doctoral; if studying at multiple levels simultaneously, we asked for the highest), and field of study (open-ended). To assess the level of students' PSA, we used a short form of the Personal Report of Confidence as a Speaker (PRCS). Originally created by Gilkinson (1942) as a 104-item paper-and-pencil measure, it was revised by Paul (1966) into a 30-item true–false format, and finally shortened by Hook et al. (2008) into a 12-item true–false format with a single factor structure. The PRCS is one of the most frequently used self-report questionnaires on PSA. The short form of the PRCS has demonstrated evidence of convergent validity with measures of anxiety, social performance anxiety, public and private self-consciousness, and shyness, and it demonstrated very good reliability (Hook et al., 2008). The short form of the PRCS includes straightforwardly worded items focusing on both physiological and emotional experiences when giving a public speech. The sum of the "true" statements results in the PSA score. A university-based language specialist assisted in translating the questionnaire into Czech. Before administering the questionnaire, we consulted with four university students, including two for whom Czech was not their first language, to assess whether the translation resulted in clear and comprehensible items.

The four students confirmed that the item wording was clear. Both English and Czech versions of the instrument are available as Supplementary Information 1. We checked the instrument's structural validity and reliability with a confirmatory factor analysis (CFA), which confirmed the instrument's single factor structure, and it showed good reliability with McDonald's $\omega_t = 0.85$. Full CFA results are available as Supplementary Information 2.

We used exclusively a Czech version of the questionnaire because most of the student body could understand Czech. The non-Czech speaking students were part of academic programs delivered entirely in English. The dynamics of PSA for students studying English programs was likely to be different, possibly mitigated by the consistency of language use in their academic environment. Therefore, we focused on those students who could understand Czech (both Czech and non-Czech) to ensure a homogenous assessment of the influence of nationality on PSA.

Sample and procedure

We conducted our research at Masaryk University in Brno, the Czech Republic, a large public university hosting many foreign students. Masaryk University was an ideal location for our study due to its emphasis on student oral presentations as a common component of teaching, assessment, and its emphasis on student-based collaborative teaching, as outlined in the University's Strategic Plan for 2021–2028 (Masaryk University, 2021). Hence, our findings should be applicable to similar higher-education contexts with public speaking being a common part of student lives. We aimed to collect data from as many Masaryk University students as possible with diverse demographic characteristics. Data collection occurred from November 3rd to 17th, 2022, in the middle of the Fall semester. Initially, we invited University students to complete the questionnaire via the University's online information system; all students logging into the system saw our call for participation as an advertisement. Simultaneously, we posted our call for participation on several Masaryk University student-run Facebook and Instagram pages and groups. After the first week of data collection, noting underrepresentation of doctoral students in our sample, we sent an email to two hundred doctoral students from all faculties at random, asking them to complete the questionnaire.

Since the items on nationality and field of study were open-ended, we coded them into two categories to allow for statistical analysis: Czech and non-Czech. Most non-Czech students were Slovak, given Masaryk University's popularity among Slovak students. Considering the historical union of the Czech Republic and Slovakia before 1993 and the similarity of the Czech and Slovak languages, we performed a sensitivity analysis with three categories of nationality – Czech, Slovak, and other – in addition to the main analysis, to control for the possibility that Slovak students might be very similar to Czech ones while differing from other nationalities. The sensitivity analysis yielded the same results as the main analysis; therefore, we report only the main analysis with two categories: Czech and non-Czech students. The results of the sensitivity analysis are available as Supplementary Information 3. We coded the field of study into three categories: social sciences and humanities (SOC-HUM), natural sciences and medicine (NAT-MED), and those who studied both of the aforementioned categories (both).

Table 1 shows the descriptive statistics of the sample compared to the University. Our sample covered 5.40% of all Masaryk University students and was diverse based on all observed demographic characteristics. We had a majority of women in terms of gender, Czech students as the majority nationality, students from vocational high schools as the predominant educational background, those pursuing bachelor's degrees as the leading level of education, and students in social sciences and humanities as the primary fields of study. We conducted a series of χ^2 tests to assess if the distribution of demographic characteristics in our sample significantly differed from that of the University, comparing the distributions of gender, nationality, study level, and field of study. Information on age and high school type, as well as non-binary persons, was not available from the University. Despite our best efforts, our sample differed in the distribution of all available demographic characteristics from the University, with all p-values from the χ^2 tests being $< .01$. Therefore, we decided to employ a weighting procedure aiming to make our sample as representative of the entire University student population as possible and to enhance the generalizability of our findings. For each student in our sample, we computed weights as the product of the ratios of population to sample proportions for each characteristic. This method ensured that individuals from under-represented groups were given higher weights while over-represented groups were assigned lower weights. We then normalized the calculated weights to ensure their sum equaled the total number of observations in our sample, preserving the overall sample size in weighted analyses. As the distribution of non-binary persons at the University was unknown, we kept their weights at 1. Following the weighting procedure, our sample was largely representative of the University population in terms of nationality, study level, and field of study.

Table 1 Descriptive statistics of the sample compared to the University

	Sample		University	
	N (%)	M (SD)	N (%)	M (SD)
All students	1745		32,335	
By age		22.36 (3.30)		unknown
By gender				
Men	476 (27.27%)		12,287 (38.00%)	
Women	1255 (71.92%)		20,048 (62.00%)	
Non-binary	14 (0.80%)		unknown	
By nationality				
Czech	1437 (82.35%)		24,183 (74.79%)	
Non-Czech	308 (17.65%)		8152 (25.21%)	
By high school type			unknown	
Vocational	1379 (79.03%)			
Academic	366 (20.97%)			
By study level[a]				
Bachelor's	1001 (57.36%)		16,320 (50.47%)	
Master's	521 (29.86%)		15,058 (46.57%)	
Doctoral	183 (10.49%)		2609 (8.07%)	
By field of study				
SOC-HUM	1174 (67.28%)		19,499 (60.30%)	
NAT-MED	499 (28.60%)		11,184 (34.59%)	
Both	72 (4.12%)		1652 (5.11%)	

[a]In the University's student population, the sum and percentages of students divided by study level do not add up to 100% of the total student body. This discrepancy occurs because the University's statistics for study levels are based on the number of active studies, not active students. Consequently, students enrolled in multiple study levels simultaneously are counted more than once, contributing to multiple study level counts

Data analysis

To test our hypotheses and assess how students' demographic characteristics predicted their PSA scores, we applied a multivariate regression analysis – a statistical technique allowing simultaneous examination of the relationship between multiple predictor variables and a dependent variable. By including several predictors in the model, we controlled for potential confounding variables. We adopted a theory-driven approach to model development, instead of relying on forward or backward selection. By grounding our model in theory, we ensured that each variable included is interpretable and relevant to the phenomenon under study. To counter the multiple hypothesis testing problem, we employed the Benjamini–Hochberg procedure (Benjamini & Hochberg, 1995), which specifically controls for the false rate discovery.

Results

Comparing students' PSA scores shows differences based on several demographic characteristics. Table 2 presents descriptive statistics, and Fig. 1 depicts plots of students' PSA scores in relation to their demographic characteristics. Our analysis found that age is negatively correlated with PSA scores. Among the gender categories, men had the lowest PSA scores, with women scoring nearly 1.5 times higher and non-binary individuals almost twice as high as men. The data showed no differences between Czech and non-Czech students. Students who graduated from academic high schools exhibited somewhat higher PSA scores compared to those from vocational high schools. Doctoral students had the lowest PSA scores, with master's and bachelor's students scoring approximately twice as high as doctoral students. Finally, students in the social sciences and humanities (SOC-HUM) displayed slightly higher PSA scores than those in natural sciences and medicine (NAT-MED) and students studying both fields.

The regression analysis suggests that the only demographic characteristics significantly predicting students' PSA scores were gender, high school, and study level. Table 3 shows the results of the regression analysis. Age was not a significant predictor of PSA after controlling for study level, hence, we do not confirm H1. On the other hand, we confirmed H2. Gender emerged as the strongest predictor among the observed characteristics, with women having significantly higher PSA than men and non-binary students exhibiting the highest PSA levels among the gender categories. We found no evidence for H3, with nationality not predicting students' PSA scores. We found an opposite direction for H4 than we expected. High school type significantly predicted PSA, but with graduates from academic high schools showing higher PSA levels than those from vocational schools. Study level was a significant predictor as well – confirming H5 – with master's students showing significantly lower PSA than bachelor's, and doctoral students having the lowest PSA among the groups. We found no evidence for H6, with field of study not predicting

Table 2 Descriptive statistics of the students' PSA scores

	M (SD)
All students	6.37 (3.51)
By gender	
Men	4.79 (3.37)
Women	6.94 (3.38)
Non-binary	8.64 (2.62)
By nationality	
Czech	6.37 (3.55)
Non-Czech	6.37 (3.33)
By high school type	
Vocational	6.23 (3.51)
Academic	6.91 (3.46)
By study level	
Bachelor's	7.05 (3.29)
Master's	6.12 (3.49)
Doctoral	3.40 (3.12)
By field of study	
SOC-HUM	6.58 (3.52)
NAT-MED	5.93 (3.37)
Both	6.10 (3.94)

students' PSA scores. Our model explained over 14% of the variability in students' PSA scores.

Discussion

Our findings contribute new perspectives to the body of research on PSA in higher education students. By confirming that several demographic characteristics significantly predict students' levels of PSA, our study identifies the most at-risk populations. This understanding is crucial for developing targeted interventions and support mechanisms tailored to the needs of these groups, aiming to enhance their academic experience and overall well-being.

We support the thesis present in some previous research that women have higher PSA levels than men (Caballo et al., 2008; Hunter et al., 2014; Marinho et al., 2017; Perveen et al., 2018). Our study adds a new dimension by including non-binary students – a group so far overlooked in PSA research and our results indicate that non-binary students experience higher PSA levels than both men and women. Building on the insights provided by stereotype threat theory (Laguía et al., 2022; Neal-Jackson, 2020; Seo & Lee, 2021), our findings suggest that the heightened PSA levels observed in women and even more so in non-binary students may be partly attributed to the pervasive influence of societal stereotypes and expectations. Women often confront stereotypes questioning their assertiveness and competence, particularly in public and professional domains. This societal backdrop may intensify the pressure and anxiety associated with public speaking, as women may fear that their performance will reinforce these negative stereotypes, leading to heightened PSA. For non-binary students, the situation seems to be compounded by additional layers of visibility and vulnerability. Non-binary individuals navigate a social landscape where their gender identity itself is often misunderstood or invalidated, placing them at an increased risk of experiencing stereotype threat not just regarding their competence, but their identity as well. The lack of representation and acknowledgment of non-binary individuals in many social spheres, including education, may further exacerbate this issue, making public speaking a potentially

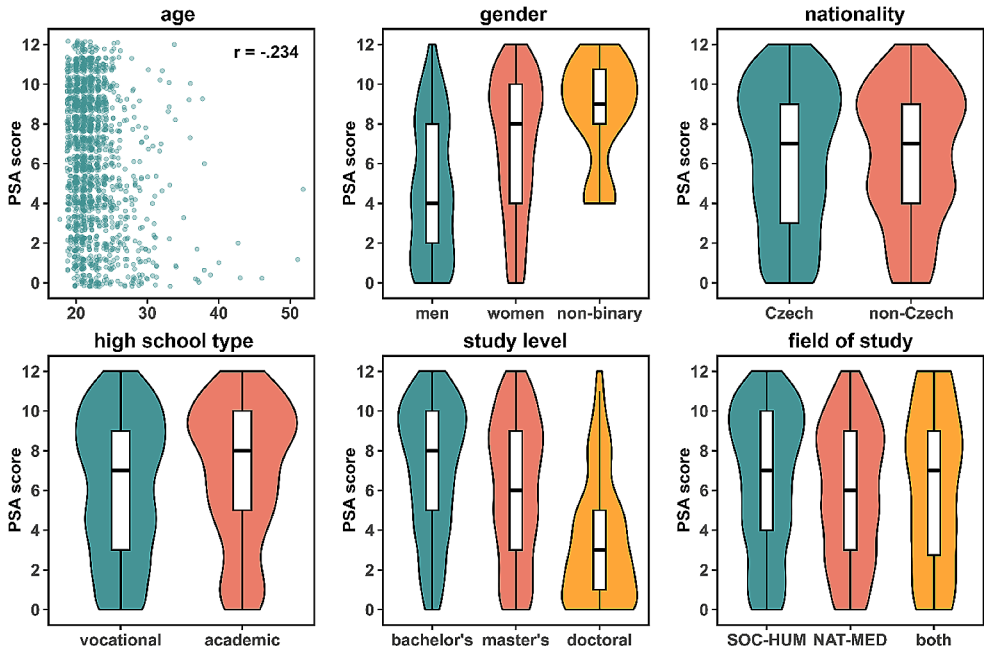

Fig. 1 Plots of the students' PSA scores related to their demographic characteristics

Table 3 Regression analysis results

	Estimate	SE	Adjusted *p*-value
Intercept	6.40	.73	< .01
H1 – age	-.04	.03	.25
H2 – gender (men as ref.)			
Women	1.80	.16	< .01
Non-binary	3.32	.97	< .01
H3 – nationality (Czech as ref.)	.27	.18	.22
H4 – academic high school (vocational as ref.)	.60	.21	.01
H5 – study level (bachelor's as ref.)			
Master's	-.75	.17	< .01
Doctoral	- 2.83	.38	< .01
H6 – field of study (SOC-HUM as ref.)			
NAT-MED	-.10	.17	.62
Both	-.04	.37	.92

Adjusted R^2: .14

more distressing experience for them compared to both men and women.

Our findings are in line with previous research, indicating that increasing age does not necessarily lead to lower levels of PSA among students (Marinho et al., 2017, 2019; Phillips et al., 1997). Contrary to expectations that older students might benefit from enhanced emotional regulation skills, our results suggest that the level of study serves as a more significant predictor of lower PSA. This highlights the pivotal role of university exposure to public speaking in mitigating anxiety, suggesting that structured opportunities to engage in public speaking within the university curriculum may have a more direct impact on reducing PSA than the gradual increase in emotional regulation typically associated with aging.

An intriguing finding of our study is the higher levels of PSA among students who graduated from academic track high schools, as opposed to those from vocational backgrounds. This contrasts with the expectation that more exposure to public speaking during high school would be related to lower PSA levels in higher education. Two reasons might account for this phenomenon. First, academic track schools often have a strong focus on academic achievement and performance, which may inadvertently heighten performance-related anxieties, including public speaking. Second, it may be a case of self-selection bias when students who choose academic tracks may inherently have different personality traits or anxiety levels compared to those who opt for vocational tracks, potentially predisposing them to higher levels of PSA. The impact of the type of high school on PSA might also vary across different educational systems and cultures. In regions where academic high schools place a greater emphasis on public speaking as part of the curriculum, the relationship between high school type and PSA might differ.

Our analysis suggests that nationality and field of study do not significantly contribute to variations in PSA among higher education students. This finding may suggest that PSA transcends cultural and disciplinary boundaries, possibly indicating that PSA is a universal experience, deeply rooted in the fear of judgment or failure in front of peers, rather than being significantly influenced by cultural or academic differences.

Limitations

While our study offers important insights into public speaking anxiety, it is important to acknowledge its limitations related to the scope of the sample and language inclusivity. We conducted our research at a single university in the Czech Republic. While this provided a rich and detailed dataset from the institution, it limits the generalizability of our results. In conducting this study through an online questionnaire, it is important to recognize potential biases that may have impacted the findings. There is a possibility of self-selection bias, wherein the individuals who chose to respond to the questionnaire might differ in their PSA levels from those who did not participate. This bias could occur if, for example, students who experience higher levels of public speaking anxiety are either more likely to respond due to a stronger engagement with the subject, or less likely due to avoidance tendencies. Our analysis included non-binary students; however, the results concerning this group should be interpreted with caution due to the relatively low number of non-binary participants in our sample. Additionally, we were unable to apply weighting for these participants, as their actual distribution within the university population is unknown. We have covered a series of theoretically relevant demographic predictors; however, we cannot rule out the possibility that we missed some crucial ones which might change our estimates. Finally, the questionnaire was administered exclusively in Czech, which potentially excluded a significant portion of the university's international students who were not proficient in the language.

Implications

Our research has implications for both practice and future research. Our findings underscore the necessity of targeted intervention and support strategies for students with higher PSA levels – women, non-binary persons, those from academic high school backgrounds, and bachelor's students. With the availability of many low-cost interventions (Lindner et al., 2019; Takac et al., 2019; Tillfors et al., 2008), it is crucial that these interventions are delivered to the most at-risk student populations. The unexpected finding that graduates from academic high schools reported higher PSA

challenges the assumption that early exposure to public speaking necessarily leads to lower PSA in higher education. This suggests a need for a reevaluation of how public speaking skills are taught at high schools. Understanding the processes by which students' demographic characteristics affect their PSA could be enriched by employing qualitative or mixed-method research designs. To understand the evolution of PSA throughout the educational lifespan, longitudinal studies tracking students from high school through higher education could provide valuable insights into how and when interventions can be most effective.

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s12144-024-06216-w.

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Data availability The data this work is based on are publicly available in a repository of Masaryk University at https://is.muni.cz/th/l60av/?studium=681417;lang=en.

Declarations

As we were working only with adult participants who gave informed consent and could withdraw at any time, we did not seek approval from an ethics committee. The questionnaire was anonymous, making it impossible to match individual answers to specific people. All procedures performed in studies involving human participants were in accordance with the ethical standards of the institutional and/or national research committee and with the 1964 Helsinki declaration and its later amendments or comparable ethical standards.

Competing interests The authors have no relevant financial or non-financial interests to disclose.

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